CUSTOMER SHOPPING BEHAVIOR ANALYSIS

1. Project Title

Customer Shopping Behavior Analysis and Interactive Business Intelligence Dashboard

2. Project Objective

The primary objective of this project was to clean, analyze, and model a transactional dataset containing 3,900 e-commerce records to uncover critical insights into customer demographics, spending patterns, product performance, and subscription behaviors.

The project aimed to:

- **Automate Data Pipelines:** Transition raw transactional logs through a structured Python cleaning and feature engineering script.
- **Perform Advanced SQL Querying:** Establish a relational database framework using PostgreSQL to compute targeted business KPIs.
- **Implement Business Intelligence:** Build an interactive Power BI dashboard to transition data insights into visual, corporate-ready strategies.
- **Optimize Marketing ROI:** Provide data-backed strategic recommendations regarding customer loyalty, subscription modeling, and discount management.

3. Project Description

Overview

The **Customer Shopping Behavior Analysis** platform is an end-to-end data engineering and business intelligence solution designed to extract meaning from 3,900 purchase entries across 18 key relational metrics. It provides a single source of truth for marketing coordinators and retail operations managers looking to optimize conversion funnels.

Repository Architecture & File Approach

The project codebase is organized into modular directories to maintain production-grade standards:

Plaintext

```
Customer_Behaviour_Analysis/
|
├── data/
|   └── shopping_behavior_updated.csv     # Raw transactional source dataset (3,900 rows)
|
├── scripts/
|   ├── data_preprocessing.py            # Automated Pandas cleaning and transformation pipeline
|   └── database_loader.py               # SQLAlchemy script to export cleaned frames to PostgreSQL
|
├── sql/
```

```
│   └── analytical_queries.sql          # Core business questions and aggregation scripts
│
├── dashboard/
│   └── Customer_Behavior_Dashboard.pbix    # Final interactive Power BI dashboard application
│
└── README.md                        # Project documentation and setup guidelines
```

1. **Data Cleaning & Engineering (data_preprocessing.py):**
    - Loaded the core dataset using pandas dataframes and standardized structural schemas.
    - Resolved data sparsity by computing category-specific median ratings to fill null instances in the Review Rating column.
    - Standardized column headers into clean snake_case layout for seamless database integration.
    - Formulated an age_group classification layer to categorize populations across targeted market demographics.
    - Handled redundancy by analyzing the correlation between discount_applied and promo_code_used, dropping the latter to optimize structural performance.
2. **Database Integration & Structured Querying (database_loader.py & analytical_queries.sql):**
    - Migrated clean data into a hosted PostgreSQL relational environment.
    - Executed precise SQL logic to determine high-spending discount consumers, shipping type efficiencies, and repeat buyer behaviors.
3. **Data Visualization (Customer_Behavior_Dashboard.pbix):**
    - Constructed an advanced data model inside Power BI Desktop using optimized DAX expressions.
    - Built a streamlined layout containing structured slicers, metric cards, category distribution charts, and demographic trends.

Technologies Used

- **Data Processing & ETL:** Python (pandas, numpy, SQLAlchemy)
- **Database Layer:** PostgreSQL
- **Business Intelligence Engine:** Power BI Desktop
- **Visualization Formats:** Plotly (via initial Python testing), Power BI DAX Engine

4. Timeline Overview

Week	Activities Planned PDF	Activities Completed PDF
Week 1	Project kickoff, understanding objectives, architecture review	Kickoff call, customer behavior business requirements definitions, logic brush-up
Week 2	Milestone 1 development & codebase structural preparation	Environment setup, Python loading testing, df.info() evaluation
Week 3	Milestone 2 implementation: Data Wrangling	Imputation of missing values via categorical medians, feature engineering of age_group
Week 4	Milestone 2 completion & script validation	SQL database connections built; schema mapped out in PostgreSQL
Week 5	Milestone 3 development: Aggregations	Formulated analytical queries for revenue metrics, subscriber cohorts, and top products
Week 6	Milestone 3 completion & Power BI Connection	Imported queried data models into Power BI; validated base values
Week 7	Milestone 4 preparation: Dashboard Design	Designed UI layout, applied theme properties, wrote DAX metrics
Week 8	Final submission & closure	Final presentation, documentation compilation, code push to GitHub

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Project alignment, initialization, and repository tracking setup	27 Nov 2025
Prototype/First Draft	Successful execution of the automated Python cleaning and pipeline scripts	10 Dec 2025

Milestone	Description	Date Achieved
Mid-Term Review	Relational schema creation and verification of primary corporate analytical queries in SQL	23 Dec 2025
Final Submission	Deployment of the interactive Power BI dashboard featuring localized metric cards and charts	16 January 2026
Presentation	Completion of the repository readme file, final metrics testing, and live presentation deck	20 Jan 2026

5b. Project Execution Details

The workflow was divided cleanly into three operational milestones:

- **The ETL Phase:** Addressed data hygiene by isolating 37 missing fields in the review column and standardizing column notation styles into a unified layout.
- **The Analytical Layer:** Drafted 10 core metrics tables to isolate vital data trends—proving that while male customers drove high raw revenue volumes ($157,890 vs $75,191), spending averages remained remarkably uniform between shipping categories and membership cohorts.
- **The Interface Layer:** Assembled a multi-tier dashboard configured with intuitive filters, allowing business users to instantly assess core categories like Clothing and Accessories.

6. Snapshots / Screenshots

(Please embed the screenshot images from your workspace repository into your final document text here to match the template format)

1. **Power BI Dashboard Overview Canvas**
2. **PostgreSQL Execution Environment and Analytical Views**
3. **Python Pipeline/Jupyter Notebook Data Processing Snippets**

7. Challenges Faced

1. **Handling Missing Review Configurations:**
    - *Challenge:* Missing values could skew overall category performance models.
    - *Resolution:* Applied a localized median imputation policy instead of global averages, safeguarding structural score integrity.
2. **Data Pipeline Structural Redundancy:**
    - *Challenge:* Redundant tracking categories (e.g., promotional flags vs discount applications) introduced unnecessary noise.

- *Resolution:* Conducted correlation checking in Python and dropped the redundant tables to keep data structures clean.

3. **Visual Representation Harmony:**
    - *Challenge:* Presenting both transaction volume metrics and gross financial values clearly without over-cluttering the UI.
    - *Resolution:* Implemented a twin column-and-bar alignment system split cleanly across separate customer counts and sales tracking targets.

8. Learnings & Skills Acquired

Technical Skills

- Relational Architecture Modeling (PostgreSQL)
- Advanced Statistical Exploratory Analysis (Python / pandas)
- Interactive Interface Design and Theme Optimization (Power BI)
- Scalable Data Processing Architectures

Soft Skills

- Technical Report Documentation
- Critical System Design Thinking
- Actionable Operational Analysis
- Agile Scope Planning and Execution

9. Testimonials from Team

"This virtual internship offered deep perspective on applying business intelligence methods directly to messy transactional fields. Constructing the pipeline from raw ingestion files into a professional executive-ready dashboard highlighted the importance of clean data management practices."

10. Conclusion

This deployment successfully delivered an enterprise-ready customer shopping behavior analysis solution. By demonstrating that *Clothing* acts as the primary category driver and identifying how consistently revenue balances across distinct age demographics, the project yields key indicators to optimize marketing channels. The architecture complies fully with analytical best practices.

11. Acknowledgements

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